PURDUE EXTENSION ENTOMOLOGY · DIAGNOSTICS LIBRARY

STUDENT DATA ENTRY *GUIDE*

Welcome. Your task is to fill in spreadsheet data for an entomology diagnostics project. The data you enter will be turned into a real, public website. This guide walks you through the spreadsheet workflow, the five sheets you'll work with, and how to submit your work.

HOW THE DATA FLOWS

```
┌──────────────────────────┐
│  YOU edit 5 spreadsheets │
└──────────────────────────┘
              │  export as CSV
              ▼
┌──────────────────────────┐
│  Instructor runs script  │   npm run import-csv
│  CSV → JSON              │
└──────────────────────────┘
              │  push to GitHub
              ▼
┌──────────────────────────┐
│  Site rebuilds on Vercel │
└──────────────────────────┘
```

You'll work in five sheets — they are **linked to each other through ID columns**. Get the IDs right and everything else falls into place.

SHEET	WHAT IT HOLDS	LINKED TO
projects	Top-level projects (e.g. *Indoor Insects*)	—
groups	Categories within a project (e.g. *Ants*)	`project_id` → projects.id
species	Individual species records	`project_id`, `group_id`
lookalikes	Pairs of species commonly confused with each other	`project_id`, `species_id`, `lookalike_id`
references	Linked publications for a species	`project_id`, `species_id`

A copy of all five sheets, pre-filled with the existing *Indoor Insects* dataset as a worked example, lives in the repo at `templates/`. Use those as your starting point.

WORKING IN THE SPREADSHEET

IDS ARE THE GLUE

Every project, group, and species needs an **`id`** — a short, lowercase, URL-safe slug like `carpenter-ant` or `bed-bugs`. The id is what shows up in the website's URL (`/p/indoor-insects/g/ants/s/carpenter-ant`), so:

- Lowercase only.
- Use hyphens (`-`), never spaces or underscores.
- No accents or special characters.
- Pick one and don't change it later — you'd break links.

The `groups` sheet has a `project_id` column that **must match exactly** an `id` in the `projects` sheet. Same for `species`: its `project_id` and `group_id` must match real entries upstream.

MULTI-VALUE CELLS (TAGS, QUICK-ID POINTS)

Two columns hold lists, not single values: `tags` and `quickId`.

Separate items with the pipe character `|` (Shift + Backslash on most US keyboards). Type the whole list on one line — no newlines inside the cell.

Example for `quickId`:

```
Single, hidden node on petiole|Even, brown-black body color|Crushed workers
smell of rotten coconut|Trails along baseboards and counters
```

That's one cell with four pipe-separated items. Each pipe becomes a new bullet on the website. Leading and trailing whitespace on each item is trimmed.

Don't use commas *between* items — commas *inside* an item (e.g. "Single, hidden node on petiole") are fine and will be preserved. Spreadsheets wrap the cell in quotes automatically when exporting to CSV.

> OLDER FORMAT STILL ACCEPTED
>
> Earlier templates used `Alt+Enter` newlines inside a cell to separate items. The importer still accepts that for backward compatibility, but pipes are now preferred: cells stay single-line, the spreadsheet is easier to read, and exports are cleaner.

LONG PROSE (HABITATNOTES, MANAGEMENT, ADDITIONALNOTES)

These are paragraph-length text fields. Just type into the cell as a single block of prose. Press `Alt+Enter` if you want a paragraph break inside the field — the site will render the break as a real paragraph.

Pipes (`|`) are *only* meaningful in `tags` and `quickId`. Anywhere else they're just regular punctuation.

IMAGES

Leave the `image` and `imageCredit` columns **blank** unless your instructor tells you otherwise. The instructor will upload photos and update those fields separately. The site automatically falls back to a striped placeholder when an image isn't set.

FILE ENCODING (UTF-8)

The templates ship as **UTF-8 with a BOM**. That's the standard Unicode encoding; the BOM (a tiny invisible marker at the start of the file) tells Excel and Numbers to open the file as UTF-8 instead of guessing a different character set and mangling characters like en-dashes (`–`), em-dashes (`–`), and accented letters.

Don't change the encoding. Specifically:

- **Excel (Windows):** open with *File → Open → Browse* (don't double-click — that path can ignore the BOM). When saving, choose *CSV UTF-8 (Comma delimited)*, not plain "CSV".
- **Excel (Mac) / Numbers:** double-clicking is fine. Pick UTF-8 if asked when exporting.
- **Google Sheets:** the safest path. Upload the template (*File → Import → Replace spreadsheet*) and download as CSV when done — Google Sheets is UTF-8 throughout.

If you see characters like `Ã–` or `â€"` where en-dashes used to be, the file has been opened in the wrong encoding. Re-open the original template instead of trying to fix the mojibake by hand.

FIELD REFERENCE

PROJECTS SHEET

One row per top-level project.

COLUMN	NOTES
`id`	URL-safe slug (`indoor-insects` , `lawn-garden`).

COLUMN	NOTES
`name`	Display name (`Indoor Insects`).
`sciName`	Scientific subtitle, often a higher-level taxon or descriptor.
`blurb`	One sentence describing the project. Shown on the homepage tile.
`pattern`	One of `diagonal` , `stipple` , `weave` , `grain` , `wing` , `web` , `hex` , `scale` .
`image`	Leave blank.
`status`	`active` for live projects; `planned` for placeholders.

The placeholder tile color is auto-derived from the `id` — no need to enter it.

GROUPS SHEET

One row per group within a project.

COLUMN	NOTES
`project_id`	Must match an id from the projects sheet.
`id`	URL-safe slug (`ants` , `bed-bugs`).
`name`	Display name.
`sciName`	Scientific name (`Formicidae`).
`blurb`	One sentence about the group.
`pattern`	Same options as projects.
`image`	Leave blank.

SPECIES SHEET

One row per species. This is where most of your work happens.

COLUMN	NOTES
`project_id` , `group_id`	Must match real upstream entries.
`id`	Slug — usually the common name lowercased and hyphenated.
`common`	Common name.
`scientific`	Scientific name, no italics (the site adds them).

COLUMN	NOTES
`family`, `order`	Taxonomic family and order.
`habitat`	The categorical filter — `In Homes`, `Stored Products`, `Around Structures`, `Structural`, etc. **Be consistent across the project** — these become the filter chips on the group page.
`range`	One-line geographic range (`Statewide`, `Southern Indiana`).
`size`	Adult body size with units (`6–13 mm`).
`tags`	Pipe-separated. Short labels (`Trail-forming|Sweet-feeding`).
`quickId`	Pipe-separated. 3–5 short identification points joined with `|`.
`habitatNotes`	Paragraph: where and when this species turns up, what signals its presence.
`management`	Paragraph: control approach. **Optional** — leave blank if you don't have notes; the section auto-hides.
`additionalNotes`	Optional free-form prose rendered at the bottom. Skip if not needed.
`image`, `imageCredit`	Leave blank.

LOOKALIKES SHEET

One row per *directional* pair. If species A is commonly confused with species B, add **two rows**: one A→B, one B→A. The site only shows lookalikes pointing outward from the species you're viewing.

COLUMN	NOTES
`project_id`	Must match a project.
`species_id`	The species you're on.
`lookalike_id`	A species id (in the same project) that's commonly confused with it.

REFERENCES SHEET

One row per reference. Optional — leave the sheet empty if a species has no linked publications.

COLUMN	NOTES
`project_id`, `species_id`	Which species this reference belongs to.
`source`	The publishing body (`Purdue Extension`).

COLUMN	NOTES
`label`	The publication title.
`url`	Direct link to the document or page.
`n`	Publication number, if any (`E-26`).

SUBMITTING YOUR WORK

When you're done editing, your instructor will tell you which of these to do:

> OPTION A — SINGLE GOOGLE SHEET
>
> Share the sheet with your instructor (View access is fine). They will export each tab to CSV and import.

> OPTION B — FIVE CSV FILES
>
> From your spreadsheet, export each sheet as a `.csv` file (in Google Sheets: *File → Download → Comma-separated values*). Send the five files to your instructor.

> OPTION C — DIRECT REPO EDIT (ADVANCED)
>
> If you have GitHub access, replace the files in `templates/` with your CSVs, run `npm run import-csv`, commit, and push. The site updates when Vercel finishes its build.

COMMON PITFALLS

- **Mismatched IDs.** A typo in `species.project_id` that doesn't appear in `projects.id` means the species silently won't render. Double-check every `*_id` column.
- **Inconsistent `habitat` values.** If three students each invent a slightly different label (`In homes` vs. `In Homes` vs. `Indoor`), they all become separate filter chips. Decide on the canonical labels with your group before you start.
- **Commas inside cells.** Spreadsheets handle this automatically when exporting to CSV — but if you're hand-writing CSV, wrap any field containing a comma in double quotes.
- **Smart quotes.** If your software auto-converts `"` to `“` or `-` to `–`, that's usually fine for prose but not for the `id` columns. Keep IDs ASCII.
- **Lost the pipe character.** A bare pipe `|` is what splits items in `tags` and `quickId`. If autocorrect rewrites it as something fancier, the importer won't recognize it. The character you

want is `Shift` + `\` on a US keyboard.

- **Mojibake (`Ã–`, `â€"`, etc.).** The file was opened in the wrong character encoding. Re-open the original UTF-8 template — see *File encoding* above.
- **Blank required fields.** `id`, `common`, `scientific`, `family`, `order`, `habitat`, `range`, `size`, `quickId`, `habitatNotes` are required.